

Pendulum Hardware Constants

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TABLE I
MEASURED AND CALCULATED CONSTANTS

Variable	Value	Method
R	16 ohms	Data Sheet
K _m	0.525 Nm/A	Calculation with Data Sheet Values
L ₁	0.1231 m	Ruler
M ₂	0.00377 kg	Scale
J ₂	0.0001374 kgm ²	Calculated
l ₂	0.1472 m	Ruler
b ₁	0.0146 Nms/rad	Solving for transfer function constants then relating them to dynamics
J ₀	0.0024 kgm ²	Solving for transfer function constants then relating them to dynamics
b ₂	4.72*10 ⁵ Nms	Measured the damping of the pendulum experimentally

$$B = \begin{bmatrix} 0 \\ 0 \\ 15.03 \\ 9.67 \end{bmatrix} \quad (2)$$

The C matrix is shown by (3).

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad (3)$$

The D matrix is shown by (4).

$$D = [0] \quad (3)$$

The impulse response of the pendulum absolute angle is shown in Fig 1 and the impulse response of the platen angle is shown in Fig 2.

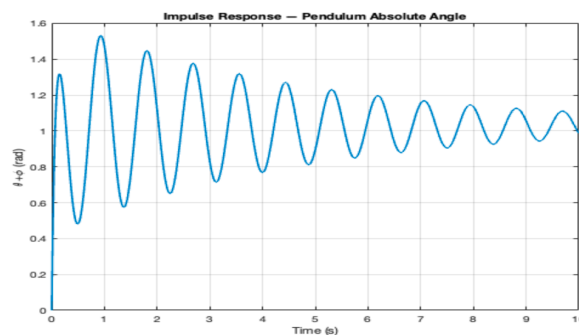


Fig 1.

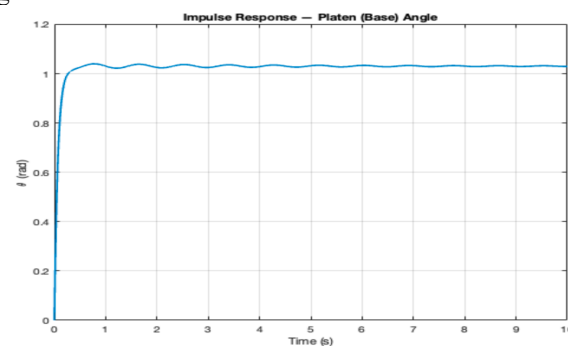


Fig 2.

State Space Matrices

The A matrix is shown by (1).

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -2.075 & -14.57 & -0.0139 \\ 0 & -52.57 & -9.38 & -0.3525 \end{bmatrix} \quad (1)$$

The B matrix is shown by (2).